SCX OOD Detection Lower Bound

SCX

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Abstract

Proves a finite-expert lower bound for out-of-distribution detection via Spring gatekeeper scores. Minimum detectable OOD shift: delta$_{m}in = sigma/sqrt(M).Falsealarmanddetectionpowerbothcontrolled$

0.1 OOD Detection Lower Bound

The CUSUM statistic above monitors temporal drift across a rollout. Spring also induces a one-sample OOD test: a sample is suspicious when its gatekeeper score is atypical relative to scores stored in the in-distribution memory. To avoid collision with the CUSUM statistic S_t, write $G_t(x)$ for the per-sample Spring gatekeeper score; this is the score denoted $S_t(x)$ in the Spring gating notes.

Let $\mathfrak{M}_t$ be the Spring memory bank of accepted in-distribution samples. For M expert modules, let $Z_{m,t}(x) \in [0,1]$ be expert m's normalized validity score for sample x at time t, and define the aggregate gatekeeper score

$$G_t(x) = \frac{1}{M}\sum_{m=1}^{M} Z_{m,t}(x). \tag{1}$$

The memory bank estimates the in-distribution score center $\mu_{\text{in},t} = \mathbb{E}[G_t(X) \mid X \sim P_{\text{in},t}]$ by $\widehat{\mu}_{\text{in},t}$.

Definition 0.1 (Spring OOD Hypothesis Test)**.** OOD detection is the binary hypothesis test

$$H_0 : X \sim P_{\text{in},t}, \qquad H_1 : X \sim P_{\text{out},t}, \tag{2}$$

with OOD shift

$$\delta_{\text{OOD}} = \left|\mathbb{E}[G_t(X) \mid H_1] - \mu_{\text{in},t}\right|. \tag{3}$$

For a tolerance $\epsilon > 0$, the ideal calibrated Spring detector is

$$\varphi_{\epsilon,t}(x) = |G_t(x) - \mu_{\text{in},t}| > \epsilon, \tag{4}$$

where $\varphi_{\epsilon,t}(x) = 1$ means "declare OOD." In deployment, $\mu_{\text{in},t}$ is replaced by $\widehat{\mu}_{\text{in},t}$ from $\mathfrak{M}_t$; the calibration error is handled in Corollary **??**.

Theorem 0.2 (SCX OOD Detection Lower Bound)**.** *Condition on the Spring memory state $\mathfrak{M}_t$. Assume that $Z_{1,t}(X), \dots, Z_{M,t}(X)$ are independent and take values in $[0,1]$. Under H_0, suppose $\mathbb{E}[G_t(X) \mid H_0] = \mu_{\text{in},t}$; under H_1, suppose $|\mathbb{E}[G_t(X) \mid H_1] - \mu_{\text{in},t}| = \delta_{\text{OOD}}$. Then, writing $p_i(\epsilon) = \mathbb{P}(\varphi_{\epsilon,t}(X) = 1 \mid H_i)$, the test in (4) satisfies*

$$p_0(\epsilon) \le 2\exp(-2M\epsilon^2), \tag{5}$$
$$p_1(\epsilon) \ge 1 - \exp\big(-2M(\delta_{\text{OOD}} - \epsilon)^2\big) \tag{6}$$

for every $0 < \epsilon < \delta_{\text{OOD}}$. In particular, if the OOD shift has margin $\delta_{\text{OOD}} \ge 2\epsilon$, then

$$\mathbb{P}(\textit{detect OOD} \mid \delta_{\text{OOD}} \ge 2\epsilon) \ge 1 - \exp(-2M\epsilon^2). \tag{7}$$

Proof. The bounds follow directly from Hoeffding's inequality applied to the M independent gatekeeper scores. $\square$